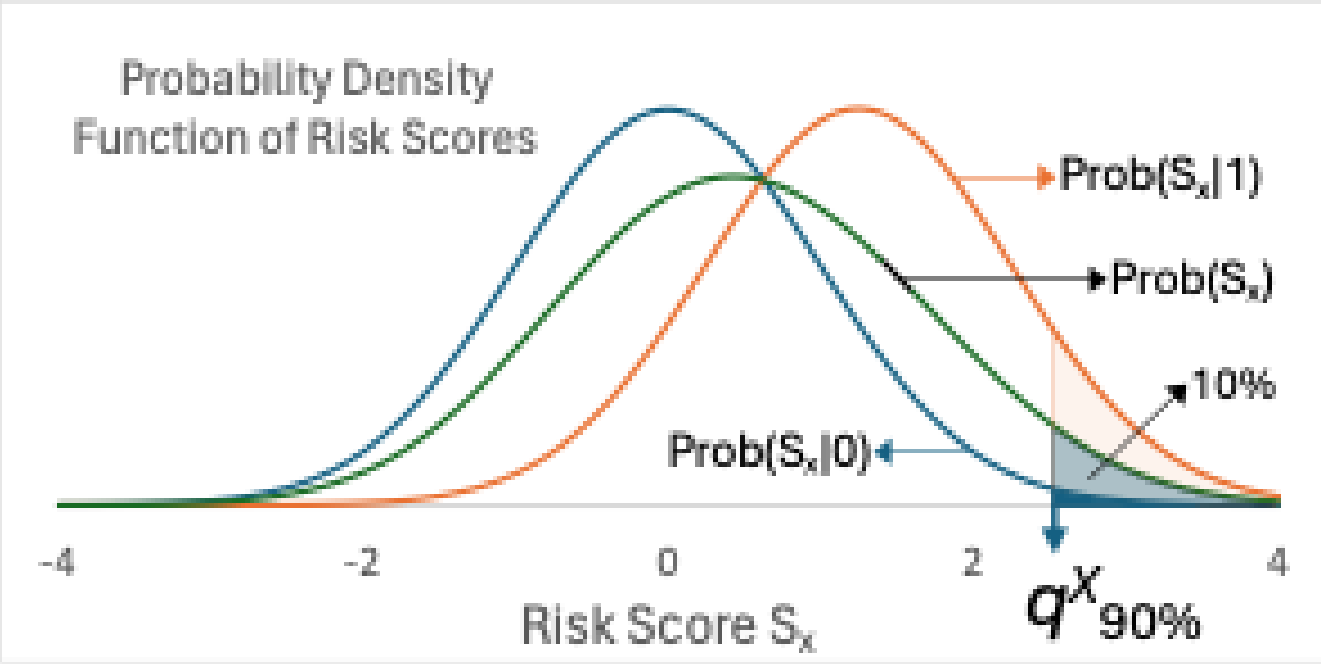
DREAMx Scoring and evaluation criteria

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Scoring the Challenge



$$OR = \frac{P(y = 1 \mid S > q)}{P(y = 0 \mid S > q)} \Big/ \frac{P(y = 1 \mid S \leq q)}{P(y = 0 \mid S \leq q)}$$

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Likelihood Function for logistic regression

$$\ell(\boldsymbol{\beta}) = \prod_{i=1}^{n} P(Y_i = y_i|\mathbf{x}_i) = \prod_{i=1}^{n} \pi^{y_i}(1-\pi)^{1-y_i}$$

where $\pi = \left(\frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}} \right)$, y_i is the phenotype for individual i, $\boldsymbol{x}_i$ is the vector of covariates for individual,

and β are estimated via maximum likelihood

Models:

1) Full Model: Covariates for the full model include: the PRS, sex, age, and genetic principal components. β are estimated from the testing scores and applied to the validation data.

2) PRS Only Model: only includes the PRS

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Scoring Metrics For Validation Phase Leaderboards

Nalgekerke R^2:

$$R^2_{\text{Nag}} = \frac{1 - \left(\frac{L_o}{L_x}\right)^{\frac{2}{N}}}{1 - (L_0)^{\frac{2}{N}}}$$

where L_0 is the likelihood of the null model and L_x is the likelihood of the full model. Measures improvement of adding the PRS to the model.

Odds Ratio: $e^{\beta_{RG}}$

Scores are ranked and grouped as top 10% (RG=1) versus lower 90% (RG=0). Measures how well the PRS distinguishes high risk individuals.

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Final Scoring

The final scoring will be conducted on the holdout dataset, for which teams must submit PRS values for all individuals.

The scoring differs from the validation phase in two respects:

- First, PRS scores and the parameters of the logistic regression models that incorporate PRS (the genetics-only model and the full model) will be trained using all individuals in the combined training and validation sets.
- Second, the odds ratio (OR) and Nagelkerke's R^2 metrics will be estimated on the holdout set using 1,000 bootstrap resamples.

To determine whether Team A outperforms Team B, we compare the primary metric (OR) from the full model across bootstrap samples.

- Team A is considered superior if its OR exceeds that of Team B in at least a 3:1 ratio of bootstrap iterations. This ratio, defined as the Bayes factor (BF), quantifies the strength of evidence in favor of Team A. If the BF for OR is 3 or greater, Team A is declared the better-performing team.

If this criterion is not met (BF < 3), the comparison proceeds to the secondary metric, Nagelkerke's R^2. The same bootstraps-based procedure is applied: if Team A achieves a BF of 3 or greater relative to Team B on R^2, Team A is deemed superior. If neither metric yields a BF of at least 3, the result is declared a tie.

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The deciphEHR PRS DREAMx Challenge

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